AMATH/PMATH 331: Applied Real Analysis — Fall 2026

Assignment 2 Submission; due Thursday, October 8, 2026

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[1] Let $(\mathbf{x}_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Write $\mathbf{x}_k = (x_{k1}, \ldots, x_{kn})$, so that $(x_{kj})_{k\in\mathbb{N}}$ is a sequence in $\mathbb{R}$ for each $j = 1, \ldots, n$.

[a] Let $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{R}^n$. Prove that $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{a}$ if and only if $\lim_{k\to\infty} x_{kj} = a_j$ for each $j = 1, \ldots, n$. (*Don't forget to prove both directions!*)

Answer:

Proof. Suppose $\lim_{k\to\infty} x_k = \boldsymbol{a}$. So $\forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ \|x_k - \boldsymbol{a}\| < \varepsilon$.
Let $\varepsilon > 0$.

$$
\begin{aligned}
&(x_{kj} - a_j)^2 \geq 0 && \forall j \in [n] \\
&(x_{kj} - a_j)^2 \leq \sum_{i=1}^{n} (x_{ki} - a_i)^2 && \forall j \in [n] \\
&|x_{kj} - a_j| \leq \sqrt{\sum_{i=1}^{n} (x_{ki} - a_i)^2} = \|x_k - \boldsymbol{a}\| < \varepsilon && \forall j \in [n], \forall k \geq N \\
&|x_{kj} - a_j| < \varepsilon && \forall j \in [n], \forall k \geq N \\
&\therefore \lim_{k\to\infty} x_{kj} = a_j && \forall j \in [n]
\end{aligned}
$$

Suppose $\forall j \in [n] : \ \lim_{k\to\infty} x_{kj} = a_j$. So $\forall j \in [n] : \ \forall \varepsilon > 0 : \ \exists N_j \in \mathbb{N} : \ \forall k \geq N_j : \ |x_{kj} - a_j| < \varepsilon$.
Let $\varepsilon' = \frac{\varepsilon}{\sqrt{n}} > 0$. Since n is finite, take $N = \max\{N_j\}$. Then, it holds that:

$$
\begin{aligned}
&|x_{kj} - a_j| < \frac{\varepsilon}{\sqrt{n}} && \forall k \geq N, \forall j \in [n] \\
&|x_{kj} - a_j|^2 < \frac{\varepsilon^2}{n} && \forall k \geq N, \forall j \in [n] \\
&\sum_{j=1}^{n} (x_{kj} - a_j)^2 < \varepsilon^2 && \forall k \geq N \\
&\|x_k - \boldsymbol{a}\| = \sqrt{\sum_{j=1}^{n} (x_{kj} - a_j)^2} < \varepsilon && \forall k \geq N \\
&\therefore \lim_{k\to\infty} x_k = \boldsymbol{a}
\end{aligned}
$$

□

[b] Prove that $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}^n$ if and only if $(x_{kj})_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$ for each $j = 1, \ldots, n$. (*Don't forget to prove both directions!*)

Answer:

Proof. Suppose $(x_k)_{k\in\mathbb{N}}$ is Cauchy. So $\forall\varepsilon > 0 : \exists N \in \mathbb{N} : \forall k, l \geq N : \|x_k - x_l\| < \varepsilon$. Let $\varepsilon > 0$.

$$
\begin{aligned}
&(x_{kj} - x_{lj})^2 \geq 0 && \forall j \in [n] \\
&(x_{kj} - x_{lj})^2 \leq \sum_{i=1}^{n} (x_{ki} - x_{li})^2 && \forall j \in [n] \\
&|x_{kj} - x_{lj}| \leq \sqrt{\sum_{i=1}^{n} (x_{ki} - x_{li})^2} = \|x_k - x_l\| < \varepsilon && \forall j \in [n], \forall k, l \geq N \\
&|x_{kj} - x_{lj}| < \varepsilon && \forall j \in [n], \forall k, l \geq N \\
&\therefore (x_{kj})_{k\in\mathbb{N}} \text{ is Cauchy} && \forall j \in [n]
\end{aligned}
$$

Suppose $\forall j \in [n] : (x_{kj})_{k\in\mathbb{N}}$ is Cauchy. So $\forall j \in [n] : \forall\varepsilon > 0 : \exists N_j \in \mathbb{N} : \forall k, l \geq N_j : |x_{kj} - x_{lj}| < \varepsilon$.
Let $\varepsilon' = \frac{\varepsilon}{\sqrt{n}} > 0$. Since n is finite, take $N = \max\{N_j\}$. Then, it holds that:

$$
\begin{aligned}
&|x_{kj} - x_{lj}| < \frac{\varepsilon}{\sqrt{n}} && \forall k, l \geq N, \forall j \in [n] \\
&|x_{kj} - x_{lj}|^2 < \frac{\varepsilon^2}{n} && \forall k, l \geq N, \forall j \in [n] \\
&\sum_{j=1}^{n} (x_{kj} - x_{lj})^2 < \varepsilon^2 && \forall k, l \geq N \\
&\|x_k - x_l\| = \sqrt{\sum_{j=1}^{n} (x_{kj} - x_{lj})^2} < \varepsilon && \forall k, l \geq N \\
&\therefore (x_k)_{k\in\mathbb{N}} \text{ is Cauchy}
\end{aligned}
$$

□

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[2] The two parts of this problem *are* related.

[a] Let $E \subseteq \mathbb{R}$ be nonempty. If E is bounded below, so that $\inf E$ exists, prove that there exists a sequence (a_k) with $a_k \in E$ for all k, and $\lim_{k\to\infty} a_k = \inf E$.

Answer:

Proof. Let $b = \inf E$.
From the definition of the infimum, we have that $\forall\varepsilon > 0 : \exists a_k \in E : b \leq a_k < b + \varepsilon$. Taking $\varepsilon = \frac{1}{k}$ for every k, we can construct the sequence (a_k) with $a_k \in E$ such that $b \leq a_k < b + \frac{1}{k}$. Since

$$\lim_{k\to\infty} \frac{1}{k} = 0 \quad \lim_{k\to\infty} b = b \quad \lim_{k\to\infty} b + \frac{1}{k} = b$$

The squeeze theorem gives $\lim_{k\to\infty} a_k = b$. □

[b] Let $A \subseteq \mathbb{R}^n$ be nonempty, and $\mathbf{x} \in \mathbb{R}^n$. We define the *distance* from A to $\mathbf{x}$ to be

$$\text{dist}_A(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}.$$

(Note that $\text{dist}_A(\mathbf{x})$ is well-defined and $\text{dist}_A(\mathbf{x}) \geq 0$, because it is the infimum of a nonempty set of nonnegative real numbers.) Prove that there exists a sequence $(\mathbf{y}_k)$ in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k and $\lim_{k\to\infty}\|\mathbf{x} - \mathbf{y}_k\| = \text{dist}_A(\mathbf{x})$.

Answer:

Proof. Let $E = \{\|\boldsymbol{x} - \boldsymbol{y}\| : \boldsymbol{y} \in A\} \subseteq \mathbb{R}$
First, lets show that $\inf E$ exists. Since A is non-empty, then E must also be non-empty. And since all the elements of E are norms of vectors, which are always non-negative, 0 is a lower bound of E. Therefore, $\inf E$ exists. We denote $\text{dist}_A(\boldsymbol{x}) = \inf E$.
From [a] we get that there exists a sequence $(a_k)_{k\in\mathbb{N}}$ with $a_k \in E$ for all k and $\lim_{k\to\infty} a_k = \text{dist}_A(\boldsymbol{x})$.
$\forall k \in \mathbb{N}$: choose $\boldsymbol{y}_k \in A$ such that $\|\boldsymbol{x} - \boldsymbol{y}_k\| = a_k$, which must exist given our definition of E.
So $(\boldsymbol{y}_k)_{k\in\mathbb{N}}$ must be a valid sequence in $\mathbb{R}^n$ with $\boldsymbol{y}_k \in A$ for all k.
Therefore, $\lim_{k\to\infty} \|\boldsymbol{x} - \boldsymbol{y}_k\| = \text{dist}_A(\boldsymbol{x})$. □

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[3] Let $A \subseteq \mathbb{R}^n$. We define the *closure* of A, denoted by $\overline{A}$, to be the subset

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \varnothing \text{ for all } \varepsilon > 0\}.$$

That is, a point $\mathbf{x} \in \mathbb{R}^n$ lies in the closure of A if and only if every open ball of any positive radius intersects A.

[a] Show that $A \subseteq \overline{A}$.

Answer:

Proof. Let $x \in A \subseteq \mathbb{R}^n$.

$$B_\varepsilon(x) = \{y \in \mathbb{R}^n : \ \|x - y\| < \varepsilon\} \qquad \forall \varepsilon > 0$$

By this deifiniton, we get that $x \in B_\varepsilon(x)$ for all $\varepsilon > 0$.
Therefore, $x \in B_\varepsilon(x) \cap A \neq \varnothing$. So $x \in \overline{A}$.
Therefore, $A \subseteq \overline{A}$. □

[b] Show that $\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \varnothing$ for any open set U containing $\mathbf{x}\}$.

Answer:

Proof. Let arbitrary $x \in \mathbb{R}^n$. We want to prove that $\forall \varepsilon > 0 : \ B_\varepsilon(x) \cap A \neq \varnothing \iff U \cap A \neq \varnothing$ for all open sets U containing x.
First, prove the forward direction:
Suppose $\forall \varepsilon > 0 : \ B_\varepsilon(x) \cap A \neq \varnothing$.
Take an arbitrary open set $U \subseteq \mathbb{R}^n$ containing x. Since U is open:

$$\exists \delta > 0 : \ B_\delta(x) \subseteq U$$

By our assumption, we know that $B_\delta(x) \cap A \neq \varnothing$. So take arbitrary $y \in B_\delta(x) \cap A$. So $y \in A$ and $y \in B_\delta(x)$. But since $B_\delta(x) \subseteq U$, we get that $y \in U$.
Therefore, $y \in U \cap A$, so $U \cap A \neq \varnothing$. Since U is arbitrary, it holds for all open sets containing x.

Now, for the reverse direction:

Suppose $U \cap A \neq \varnothing$ for all open sets U containing x.
Let arbitrary $\varepsilon > 0$ and take the open set $U = B_\varepsilon(x)$ containing x. Then, we have:

$$\forall \varepsilon > 0 : B_\varepsilon(x) \cap A \neq \varnothing$$

Therefore, we have equivalence of the 2 statements. So:

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \varnothing \text{ for all } \varepsilon > 0\} \iff$$
$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \varnothing \text{ for any open set } U \text{ containing } \mathbf{x}\}$$

□

[c] Show that $\overline{A}$ is closed.

Hint: First show there exists ε_0 such that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \varnothing$, and then show that $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \varnothing$. You will need to use part [b].

Answer:

Proof. Let's look at $\overline{A}^c = \mathbb{R}^n \setminus \overline{A}$.

$$\forall x \in \overline{A}^c : \ \exists \varepsilon > 0 : \ B_\varepsilon(x) \cap A = \varnothing$$

Take arbitrary $x \in \overline{A}^c$. Then ε_0 exists such that $B_{\varepsilon_0}(x) \cap A = \varnothing$.
Now assume for the sake of contradiction that there exists y such that

$$y \in B_{\varepsilon_0}(x) \cap \overline{A}$$

$B_{\varepsilon_0}(x)$ is an open set containing y and $y \in \overline{A}$, so $B_{\varepsilon_0}(x) \cap A \neq \varnothing$, which is a contradiction.
Therefore, $B_{\varepsilon_0}(x) \cap \overline{A} = \varnothing$. Since x is arbitrary, it holds that

$$\forall x \in \overline{A}^c : \ \exists \varepsilon > 0 : \ B_\varepsilon(x) \cap \overline{A} = \varnothing$$
$$\forall x \in \overline{A}^c : \ \exists \varepsilon > 0 : \ B_\varepsilon(x) \subseteq (\mathbb{R}^n \setminus \overline{A})$$
$$\forall x \in \overline{A}^c : \ \exists \varepsilon > 0 : \ B_\varepsilon(x) \subseteq \overline{A}^c$$
$$\therefore \overline{A}^c \text{ is open}$$

Therefore, $\overline{A}$ is closed. □

[d] If A is closed, show that $\overline{A} = A$.

Answer:

Proof. Assume A is closed. We know that $A \subseteq \overline{A}$, so we have to show $\overline{A} \subseteq A$.
Take arbitrary $y \in \overline{A}$.

$$\forall \varepsilon > 0 : \ B_\varepsilon(y) \cap A \neq \varnothing$$

We know A^c is open:

$$\forall x \in A^c : \ \exists \varepsilon > 0 : B_\varepsilon(x) \subseteq A^c$$

Which is equivalent to:

$$\forall x \in A^c : \ \exists \varepsilon > 0 : B_\varepsilon(x) \cap A = \varnothing$$

Now, assume for the sake of contradiction that $y \in A^c$. Since A^c is open and $y \in A^c$, there exists $\varepsilon_0 > 0$ such that

$$B_{\varepsilon_0}(y) \subseteq A^c$$
$$B_{\varepsilon_0}(y) \cap A = \varnothing.$$

But since $y \in \overline{A}$

$$B_{\varepsilon_0}(y) \cap A \neq \varnothing$$

a contradiction. Therefore,

$$y \notin A^c$$
$$y \in A$$

Since y is arbitrary, it holds for all $y \in \overline{A}$. Therefore, $\overline{A} \subseteq A$. □

(Combining parts [c] and [d] shows that A is closed if and only if $\overline{A} = A$.)

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[4] Let $E \subseteq \mathbb{R}^n$ be nonempty.

[a] If $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is a sequence such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$, prove that $\mathbf{p} \in \overline{E}$.

Answer:

Proof. Take arbitrary $\varepsilon > 0$.

$$B_\varepsilon(\boldsymbol{p}) = \{x \in \mathbb{R}^n : \ \|\boldsymbol{p} - x\| < \varepsilon\}$$

Since $\lim_{k\to\infty} x_k = \boldsymbol{p}$, then $\exists N \in \mathbb{N} : \ \forall k \geq N : \ \|x_k - \boldsymbol{p}\| < \varepsilon$
That is, $\forall k \geq N : \ x_k \in B_\varepsilon(\boldsymbol{p})$. And since $x_k \in E$ for all k, we have that

$$B_\varepsilon(\boldsymbol{p}) \cap E \neq \varnothing$$

Since ε was arbitrary, it holds that

$$\forall \varepsilon > 0 : \ B_\varepsilon(\boldsymbol{p}) \cap E \neq \varnothing$$

Which is exactly the condition for being a member of $\overline{E}$,
Therefore, $\boldsymbol{p} \in \overline{E}$. □

[b] Let $\mathbf{p} \in \overline{E}$. Prove that there exists a sequence $(\mathbf{x}_k)_{k\in\mathbb{N}}$ such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$. *Hint:* Consider $B_{\frac{1}{k}}(\mathbf{p})$.

Answer:

Proof. Since $\boldsymbol{p} \in \overline{E}$

$$\forall \varepsilon > 0 : \ B_\varepsilon(\boldsymbol{p}) \cap E \neq \varnothing$$

Consider the values $\varepsilon = \frac{1}{k}$ for all $k \in \mathbb{N}$. We know that for each of these values, there is at least one element both in E and $B_{\frac{1}{k}}(\boldsymbol{p})$. Call that element x_k.

More formally:

$$\forall k \in \mathbb{N} : \; x_k \in B_{\frac{1}{k}}(\boldsymbol{p}) \wedge x_k \in E$$

This is a sequence $(x_k)_{k\in\mathbb{N}}$ with $x_k \in E$ for every $k \in \mathbb{N}$.
Also, since $x_k \in B_{\frac{1}{k}}(\boldsymbol{p})$, we have that $\|x_k - \boldsymbol{p}\| < \frac{1}{k}$ for all k.
Take arbitrary $\varepsilon > 0$. And choose N such that $N > \frac{1}{\varepsilon}$.

$$\begin{aligned}
&\forall k \in \mathbb{N} : \; \|x_k - \boldsymbol{p}\| < \frac{1}{k} \\
&\forall k, l \in \mathbb{N} : \; \|x_{k+l} - \boldsymbol{p}\| < \frac{1}{k+l} < \frac{1}{k} \\
&\forall k \geq N : \; \|x_k - \boldsymbol{p}\| < \frac{1}{k} \leq \frac{1}{N} < \varepsilon
\end{aligned}$$

Since ε was arbitrary, $\lim_{k\to\infty} x_k = \boldsymbol{p}$. □

Taken together, these two statements say that $\overline{E}$ is exactly the set of limits of convergent sequences whose elements all lie in E.

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[5] Let $(\mathbf{x}_m)_{m\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Define a new sequence $(\mathbf{s}_k)_{k\in\mathbb{N}}$ by

$$\mathbf{s}_k = \sum_{m=1}^{k} \mathbf{x}_m.$$

If the sequence $(\mathbf{s}_k)$ converges, then we say that the *series* $\sum_{m=1}^{\infty} \mathbf{x}_m$ *converges*. If $\lim_{k\to\infty} \mathbf{s}_k = \mathbf{s}$, then we write $\sum_{m=1}^{\infty} \mathbf{x}_m = \mathbf{s}$.

[a] Prove that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges iff: for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon$$

for all $\ell \geq k \geq N$.
Hint: A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy.

Answer:

Proof.

$$\begin{aligned}
&\forall \varepsilon > 0 : \; \exists N \in \mathbb{N} : \; \forall l \geq k \geq N : \; \left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon \\
\iff &\forall \varepsilon > 0 : \; \exists N \in \mathbb{N} : \; \forall l \geq k \geq N : \; \left\| \sum_{m=1}^{\ell} \mathbf{x}_m - \sum_{m=1}^{k} \mathbf{x}_m \right\| < \varepsilon \\
\iff &\forall \varepsilon > 0 : \; \exists N \in \mathbb{N} : \; \forall l \geq k \geq N : \; \|s_\ell - s_k\| < \varepsilon \\
\iff &(s_k)_{k\in\mathbb{N}} \text{ is Cauchy} \\
\iff &(s_k)_{k\in\mathbb{N}} \text{ converges.}
\end{aligned}$$

□

[b] Suppose that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges. For any $N \in \mathbb{N}$, prove that $\sum_{m=N}^{\infty} \mathbf{x}_m$ converges, and that

$$\lim_{N\to\infty} \left(\sum_{m=N}^{\infty} \mathbf{x}_m \right) = 0.$$

Hint: Consider the sequence $(\tilde{\mathbf{s}}_\ell)_{\ell \geq N}$ given by $\tilde{\mathbf{s}}_\ell = \sum_{m=N}^{\ell} \mathbf{x}_m$.

Answer:

Proof. Consider the sequence $(\tilde{s}_\ell)_{\ell \geq N}$ given by $\tilde{s}_\ell = \sum_{m=N}^{\ell} x_m$.
Since we know s_ℓ converges, let $\lim_{\ell\to\infty} s_\ell = \sum_{m=1}^{\infty} x_m = p$.
Also take arbitrary $\varepsilon > 0$.

$$\exists L \in \mathbb{N} : \ \forall \ell \geq L : \ \|s_\ell - p\| < \varepsilon$$

Now,

$$\forall \ell \geq L : \ \|\tilde{s}_\ell - (p - s_{N-1})\| = \|s_\ell - p\| < \varepsilon$$

Therefore, $(\tilde{s}_\ell)_{l \geq N}$ converges to $p - s_{N-1}$.

$$\sum_{m=N}^{\infty} x_m = p - s_{N-1}$$

Let arbitrary $\varepsilon > 0$. Since $s_k \to p$, there exists $K \in \mathbb{N}$ such that

$$\forall k \geq K : \ \|s_k - p\| < \varepsilon.$$

Take $N \geq K + 1$. Then $N - 1 \geq K$, and therefore

$$\left\| \sum_{m=N}^{\infty} x_m - 0 \right\| = \|p - s_{N-1}\| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary,

$$\lim_{N\to\infty} \left(\sum_{m=N}^{\infty} x_m \right) = 0.$$

□

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[6] Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Given $\boldsymbol{x} \in \mathbb{R}^n$, prove that (x_k) has a *subsequence* converging to $\boldsymbol{x}$ if and only if, for all $\varepsilon > 0$ the open ball $B_\varepsilon(\boldsymbol{x})$ contains x_k for infinitely many k.

Answer:

Proof. Let $f : \mathbb{N} \to \mathbb{N}$ strictly increasing. And denote $(x_{f(k)})_{k\in\mathbb{N}}$ a subsequence of (x_k).
Firstly, prove:

$$\left(\lim_{k\to\infty} x_{f(k)} = \boldsymbol{x} \right) \implies \left(\forall \varepsilon > 0 : \ x_k \in B_\varepsilon(\boldsymbol{x}) \text{ for infinitely many } k \right)$$

Assume $\lim_{k\to\infty} x_{f(k)} = \boldsymbol{x}$. Let arbitrary $\varepsilon > 0$. Take $K \in \mathbb{N}$ such that $\forall k \geq K : \left\| \boldsymbol{x} - x_{f(k)} \right\| < \varepsilon$.

$$\forall k \geq K : \ x_{f(k)} \in B_\varepsilon(\boldsymbol{x})$$

We know from the archimedean property that there are an infinite amount of $k \geq K$, and since f is strictly incresing, all $f(k)$ are distinct, so there is an infinite amount of $x_{f(k)}$ such that $k \geq K$.

Since ε was arbitrary, it holds for all $\varepsilon > 0$.

Secondly, prove:

$$\Big(\forall \varepsilon > 0 : \ x_k \in B_\varepsilon(\boldsymbol{x}) \text{ for infinitely many } k\Big) \implies \big(\lim_{k\to\infty} x_{f(k)} = \boldsymbol{x}\big)$$

Assume $\forall \varepsilon > 0 : \ x_k \in B_\varepsilon(\boldsymbol{x})$ for infinitely many k.

Now, for every $j \in \mathbb{N}$ do the following:

- Apply the assumption with $\varepsilon = \frac{1}{j}$: $x_k \in B_{\frac{1}{j}}(\boldsymbol{x})$ for infinitely many k.
- Choose $f(j) = k_j$, such that:
  - $x_{k_j} \in B_{\frac{1}{j}}(\boldsymbol{x})$.
  - For $j > 1$: $f(j) > f(j-1)$ (this can always be satisfied, as for every j we have an infinite amount of k_j to choose from)

With this construction of $f(k)$, we have:

$$\forall k \in \mathbb{N} : \ \big\|x_{f(k)} - \boldsymbol{x}\big\| < \frac{1}{k}$$

$\lim_{k\to\infty} \frac{1}{k} = 0$, therefore $\lim_{k\to\infty} x_{f(k)} = \boldsymbol{x}$ □

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[7] We know that a nonconvergent sequence can have subsequences which can converge to *different* limits. Give an example of a sequence in $\mathbb{R}^n$ which has *no convergent subsequences* at all. Justify your answer. *Hint:* Recall the Bolzano-Weierstrass Theorem

Answer:

Take the sequence $(x_k)_{k\in\mathbb{N}}$ in $\mathbb{R}^n$ with $x_k = (k, 0, \ldots, 0) \in \mathbb{R}^n$.

Proof. Any subsequence of (x_k) would be of the form $(x_{f(k)})$ with $f(k)$ strictly increasing and $f(k) \geq k$.

But then $x_{f(k)} = (f(k), 0, \ldots, 0)$.

Let $x_{f(k)1} = f(k)$. Since $f(k) \geq k$ and incresing, we get $\lim_{k\to\infty} x_{f(k)1} = \lim_{k\to\infty} f(k) = \infty$. And on problem [1] of this assignment we've shown that a sequence converges iff all of it's components converge.

Therefore, $x_{f(k)}$ does not converge. And this is true for all subsequences of (x_k). □

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[8] Recall that a subset K of $\mathbb{R}^n$ is called *compact* iff every sequence in K admits a subsequence that converges to a point in K.

Let $K_1, K_2, \ldots, K_m$ be a finite collection of compact subsets of $\mathbb{R}^n$. Give two different proofs that the union $\bigcup_{j=1}^m K_j$ is compact:

[a] Using the definition of compactness.

Answer:

Proof. Denote $K = \bigcup_{j=1}^{m} K_j$.
Take an arbitrary sequence $(x_k)_{k\in\mathbb{N}}$ in K. Every element of this sequence must be in at least one of K_j, and since there are an infinite amount of elements in the sequence, there must exist at least one K_j that has an infinite amount of elements of (x_k).
We can take exactly these elements to form a subsequence $(x_{f_j(k)})$ such that $x_{f_j(k)} \in K_j$.
Since K_j is compact, we know by definition that there exists a convergent subsequence $(x_{f_j(h(k))})$ such that $\lim_{k\to\infty} x_{f_j(h(k))} \in K_j$.
Therefore, $\lim_{k\to\infty} x_{f_j(h(k))} \in K$. So for an arbitrary sequence in K we've found a subsequence that converges to a point in K. Therefore, K is compact. □

[b] Using the Heine-Borel theorem

Answer:

Proof. Denote $K = \bigcup_{j=1}^{m} K_j$.
We have that K_j is closed and bounded, for all $j \in [m]$. Let $M_1, M_2, \ldots, M_m$ be bounds for $K_1, K_2, \ldots, K_m$ respectively. Then,

$$M = \max_{j\in[m]}\{M_j\}$$

is a bound for K. So K is bounded.
Now we have to show that the union of multiple closed sets is also closed.

$$K = \bigcup_{j=1}^{m} K_j = (\bigcap_{j=1}^{m} K_j^c)^c$$

We know that K_j^c is open, and we have to show that $K^c = \bigcap_{j=1}^{m} K_j^c$ is also open.

$$\forall x \in K_j^c : \ \exists \varepsilon_j > 0 : \ B_{\varepsilon_j}(x) \subseteq K_j^c$$

Take arbitrary $x \in K^c$ and choose $\varepsilon = \min_{j\in[m]}\{\varepsilon_j\}$.

$$\begin{aligned}
&\forall y \in B_\varepsilon(x) : \ \|y - x\| < \varepsilon && \forall j \in [m] : \ \varepsilon \le \varepsilon_j \\
&\forall j \in [m] : \ \forall y \in B_\varepsilon(x) : \ \|y - x\| < \varepsilon_j \\
&\forall j \in [m] : \ \forall y \in B_\varepsilon(x) : \ y \in B_{\varepsilon_j}(x) \\
&\forall j \in [m] : \ B_\varepsilon(x) \subseteq B_{\varepsilon_j}(x) && B_{\varepsilon_j}(x) \subseteq K_j^c \\
&\forall j \in [m] : \ B_\varepsilon(x) \subseteq K_j^c \\
&B_\varepsilon(x) \subseteq \bigcap_{j=1}^{m} K_j^c = K^c
\end{aligned}$$

Since x was arbitrary, it holds for all $x \in K^c$, therefore K^c is open, and $Kisclosed$.
Since K is closed and bounded, it follows from the Heine-Borel Theorem that K is compact. □

[9] Let $E \subseteq \mathbb{R}^n$ be nonempty. For $x \in \mathbb{R}^n$, we define the *distance* from E to x, denoted $\text{dist}_E(x)$, to be

$$\text{dist}_E(x) = \inf\{\|x - z\| : z \in E\}$$

Thus we can think of dist_E as a function from $\mathbb{R}^n$ to $\mathbb{R}$.

[a] Prove that $|\text{dist}_E(x) - \text{dist}_E(y)| \leq \|x - y\|$ for all $x, y \in \mathbb{R}^n$.

Answer:

Proof. Denote $D_E(x) = \{\|x - z\| : \ z \in E\}$ for $x \in \mathbb{R}^n$. Also, $\text{dist}_E(x) = \inf D_E(x)$. Let arbitrary $z \in E$.

$$\begin{aligned} \|x - z\| \in D_E(x) &\implies \|x - z\| \geq \text{dist}_E(x) \\ \|y - z\| \in D_E(y) &\implies \|y - z\| \geq \text{dist}_E(y) \end{aligned}$$

From the triangle inequality:

$$\begin{aligned} \|x - z\| = \|(x - y) + (y - z)\| &\leq \|x - y\| + \|y - z\| \\ \text{dist}_E(x) &\leq \|x - y\| + \|y - z\| \\ \|y - z\| &\geq \text{dist}_E(x) - \|x - y\| \end{aligned}$$

Therefore, since this holds for every $z \in E$, $\text{dist}_E(x) - \|x - y\|$ is a lower bound of $D_E(y)$.
So $\text{dist}_E(y) \geq \text{dist}_E(x) - \|x - y\|$, since it is the infimum. Therefore,

$$\text{dist}_E(x) - \text{dist}_E(y) \leq \|x - y\|$$

Since x and y arbitrary, we can swap them around to get

$$\text{dist}_E(y) - \text{dist}_E(x) \leq \|x - y\|$$

Therefore,

$$|\text{dist}_E(x) - \text{dist}_E(y)| \leq \|x - y\|$$

□

[b] Prove that $\text{dist}_E : \ \mathbb{R}^n \to \mathbb{R}$ is continuous on $\mathbb{R}^n$.

Answer:

Proof. Let arbitrary $p \in \mathbb{R}^n$. Let arbitrary $\varepsilon > 0$. Take $\delta = \varepsilon$. Let arbitrary $x \in B_\delta(p)$. We need to show that $|\text{dist}_E(x) - \text{dist}_E(p)| < \varepsilon$.
From [a] we have that

$$|\text{dist}_E(x) - \text{dist}_E(p)| \leq \|x - p\|$$

And from our choice of x we have

$$\|x - p\| < \delta = \varepsilon$$

Therefore,

$$|\text{dist}_E(x) - \text{dist}_E(p)| < \varepsilon$$

Since, x and ε were arbitrary, it holds that:

$$\forall \varepsilon > 0 : \ \exists \delta > 0 : \ \forall x \in B_\delta(p) \cap \mathbb{R}^n : \ \|\text{dist}_E(x) - \text{dist}_E(p)\| < \varepsilon$$

Therefore, dist_E is continuous at p. Since p is arbitrary, dist_E is continuous on $\mathbb{R}^n$. □

[10] Let $f:\ \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that f is not continuous at x for any $x \in \mathbb{R}$.

Hint: Find a way to use the sequential characterization of continuity.

Answer:

Proof. Let arbitrary $p \in \mathbb{R} \setminus \mathbb{Q}$. So $f(p) = 1$. We need to construct a sequence (x_k) in $\mathbb{R}$ that converges to p, but the sequence $(f(x_k))$ does not converge to $f(p)$.

Using the floor function we can define the following sequence:

$$x_k = \frac{\lfloor 10^k p \rfloor}{10^k}$$

Or, a description of the sequence can be "x_k = The first k digits of the decimal representation of p".

The floor function always produces integeres, so this is the division of 2 integers, therefore $x_k \in \mathbb{Q}$ for every $k \in \mathbb{N}$.

Now, to show that (x_k) converges to p we will use the following property of the floor function:

$$\begin{aligned} &\lfloor p \rfloor \leq p < \lfloor p \rfloor + 1 \\ &\lfloor 10^k p \rfloor \leq 10^k p < \lfloor 10^k p \rfloor + 1 \\ &\frac{\lfloor 10^k p \rfloor}{10^k} \leq p < \frac{\lfloor 10^k p \rfloor}{10^k} + \frac{1}{10^k} \\ &0 \leq p - \frac{\lfloor 10^k p \rfloor}{10^k} < \frac{1}{10^k} \end{aligned}$$

So $|x_k - p| < \frac{1}{10^k}$ Let arbitrary $\varepsilon > 0$. Take $N \in \mathbb{N}$ such that $N > log_{10}(\frac{1}{\varepsilon})$.

$$\forall k \geq N:\ |x_k - p| < \frac{1}{10^k} \leq \frac{1}{10^N} < \frac{1}{10^{log_{10}(\frac{1}{\varepsilon})}} = \varepsilon$$

Therefore, (x_k) converges to p.

Since $x_k \in \mathbb{Q}$ for all k, $f(x_k) = 0$ for all k. So the sequence $f(x_k)$ converges to 0.

Therefore, we have found a sequence that converges to p, but it s image over f does not converge to the image of p. By the Sequential characterization of continuity, f is not continuous at p.

Therefore, f is not continuousat all irrational points.

Let arbitrary $p \in \mathbb{Q}$. So $f(p) = 0$. Consider the sequence $(x_k)_{k \in \mathbb{N}}$ in $\mathbb{R} \setminus \mathbb{Q}$:

$$x_k = p + \frac{\sqrt{2}}{k}$$

Since $\sqrt{2}$ is irrational, x_k is irrational for all k.

Let arbitrary $\varepsilon > 0$. Take $N \in \mathbb{N}$ such that $N > \frac{\sqrt{2}}{\varepsilon}$.

$$\forall k \geq N : \; x_k - p = \frac{\sqrt{2}}{k} \leq \frac{\sqrt{2}}{N} < \frac{\sqrt{2}}{\frac{\sqrt{2}}{\varepsilon}} = \varepsilon$$

So x_k converges to p. But since $x_k \in \mathbb{R} \setminus \mathbb{Q}$ for all k, we have that $f(x_k) = 1$ for all k. So $(f(x_k))$ converges to 1.

Therefore, we have found a sequence that converges to p, but it s image over f does not converge to the image of p. By the Sequential characterization of continuity, f is not continuous at p.

Therefore, f is not continuousat all rational points.

Therefore, f is not continuous at x for any $x \in \mathbb{R}$. $\square$